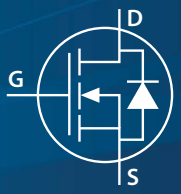
EPC2071 – Enhancement Mode Power Transistor

V_{DS}, 100 V
$R_{DS(on)}$, 2.2 mΩ max
I_D, 64 A



EPC EFFICIENT POWER CONVERSION

RoHS Pb Halogen-Free

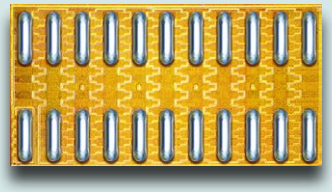
Gallium Nitride's exceptionally high electron mobility and low temperature coefficient allows very low $R_{DS(on)}$, while its lateral device structure and majority carrier diode provide exceptionally low Q_G and zero Q_{RR}. The end result is a device that can handle tasks where very high switching frequency, and low on-time are beneficial as well as those where on-state losses dominate.

Application Notes:

- Easy-to-use and reliable gate, Gate Drive ON = 5 V typical, OFF = 0 V (negative voltage not needed)
- Top of FET is electrically connected to source

Questions:

ASK AN EXPERT GaN



EPC2071 eGaN® FETs are supplied only in passivated die form with solder bars.
Die Size: 4.45 x 2.3 mm

Applications

- 48 V DC-DC Converters
- BLDC Motor Drives
- Sync Rectification for AC/DC and DC-DC
- Point of Load Converters
- Solar Converters
- Lidar
- eMobility

Benefits

- Ultra High Efficiency
- No Reverse Recovery
- Ultra Low Q_G, Q_{GD}, Q_{OSS}
- Ultra Low $R_{DS(on)}$
- Ultra Small Footprint

Maximum Ratings			
PARAMETER		VALUE	UNIT
V_{DS}	Drain-to-Source Voltage (Continuous)	100	V
	Drain-to-Source Voltage (up to 10,000 5 ms pulses at 150°C)	120	
I_D	Continuous (T_A = 25°C)	64	A
	Pulsed (25°C, T_{PULSE} = 300 µs)	350	
V_{GS}	Gate-to-Source Voltage	6	V
	Gate-to-Source Voltage	-4	
T_J	Operating Temperature	-40 to 150	°C
T_{STG}	Storage Temperature	-40 to 150	

Thermal Characteristics			
PARAMETER		TYP	UNIT
$R_{\theta JC}$	Thermal Resistance, Junction-to-Case (Case TOP)	0.4	°C/W
$R_{\theta JB}$	Thermal Resistance, Junction-to-Board (Case BOTTOM)	1.5	
$R_{\theta JA_JEDEC}$	Thermal Resistance, Junction-to-Ambient (using JEDEC 51-2 PCB)	59	
$R_{\theta JA_EVB}$	Thermal Resistance, Junction-to-Ambient (using EPC90122 EVB)	31	

Static Characteristics (T_J = 25°C unless otherwise stated)						
PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
BV_{DSS}	Drain-to-Source Voltage	V_{GS} = 0 V, I_D = 0.15 mA	100			V
I_{DSS}	Drain-Source Leakage	V_{GS} = 0 V, V_{DS} = 80 V		0.001	0.12	mA
I_{GSS}	Gate-to-Source Forward Leakage	V_{GS} = 5 V		0.03	3.2	
	Gate-to-Source Forward Leakage#	V_{GS} = 5 V, T_J = 125°C		0.3	7.1	
	Gate-to-Source Reverse Leakage	V_{GS} = -4 V		0.006	0.17	
$V_{GS(TH)}$	Gate Threshold Voltage	V_{DS} = V_{GS}, I_D = 13 mA	0.7	1.3	2.5	V
$R_{DS(on)}$	Drain-Source On Resistance	V_{GS} = 5 V, I_D = 30 A		1.7	2.2	mΩ
V_{SD}	Source-Drain Forward Voltage#	I_S = 0.5 A, V_{GS} = 0 V		1.5		V

Defined by design. Not subject to production test.

Dynamic Characteristics# (T_J = 25°C unless otherwise stated)						
PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
C_{ISS}	Input Capacitance	V_{DS} = 50 V, V_{GS} = 0 V		2664	3931	pF
C_{RSS}	Reverse Transfer Capacitance			5.4		
C_{OSS}	Output Capacitance			878	976	
$C_{OSS(ER)}$	Effective Output Capacitance, Energy Related (Note 1)	V_{DS} = 0 to 50 V, V_{GS} = 0 V		1058		
$C_{OSS(TR)}$	Effective Output Capacitance, Time Related (Note 2)			1422		
R_G	Gate Resistance			0.3		Ω
Q_G	Total Gate Charge	V_{DS} = 50 V, V_{GS} = 5 V, I_D = 30 A		18	26	nC
Q_{GS}	Gate-to-Source Charge	V_{DS} = 50 V, I_D = 30 A		6.0		
Q_{GD}	Gate-to-Drain Charge			1.8		
$Q_{G(TH)}$	Gate Charge at Threshold			4.5		
Q_{OSS}	Output Charge	V_{DS} = 50 V, V_{GS} = 0 V		71	82	
Q_{RR}	Source-Drain Recovery Charge			0		

Defined by design. Not subject to production test.
All measurements were done with substrate connected to source.
Note 1: $C_{OSS(ER)}$ is a fixed capacitance that gives the same stored energy as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS}.
Note 2: $C_{OSS(TR)}$ is a fixed capacitance that gives the same charging time as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS}.

Figure 1: Typical Output Characteristics at 25°C

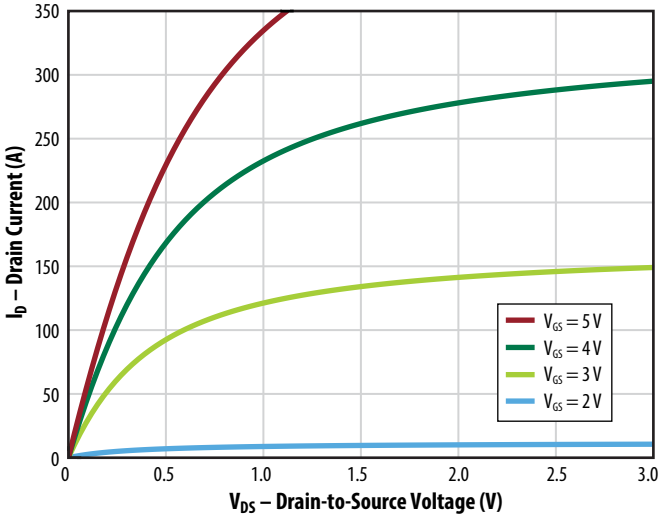


Figure 2: Typical Transfer Characteristics

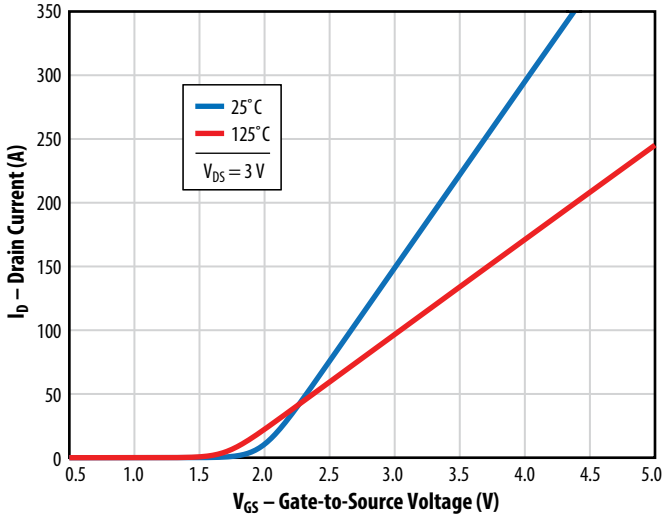


Figure 3: $R_{DS(on)}$ vs. V_{GS} for Various Drain Currents

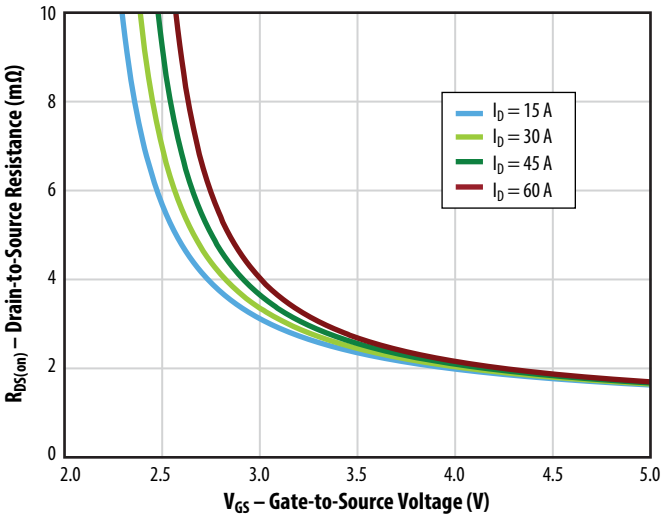


Figure 4: $R_{DS(on)}$ vs. V_{GS} for Various Temperatures

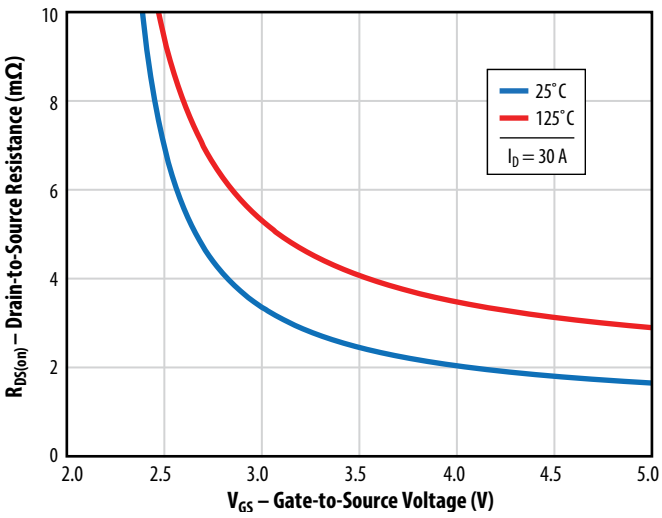


Figure 5a: Typical Capacitance (Linear Scale)

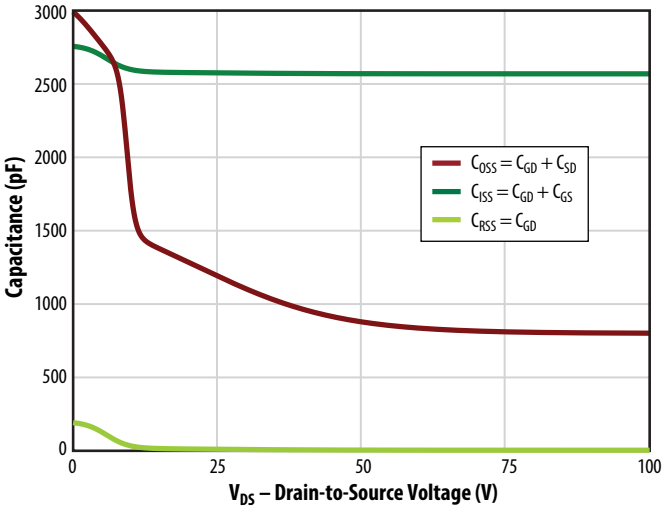


Figure 5b: Typical Capacitance (Log Scale)

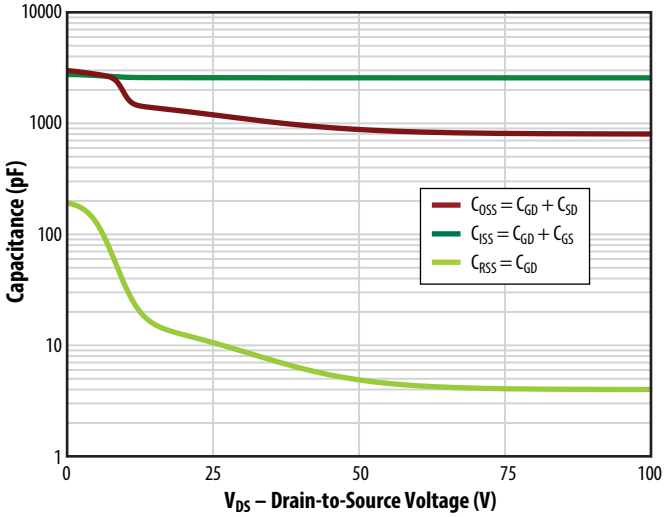


Figure 6: Typical Output Charge and C_{OSS} Stored Energy

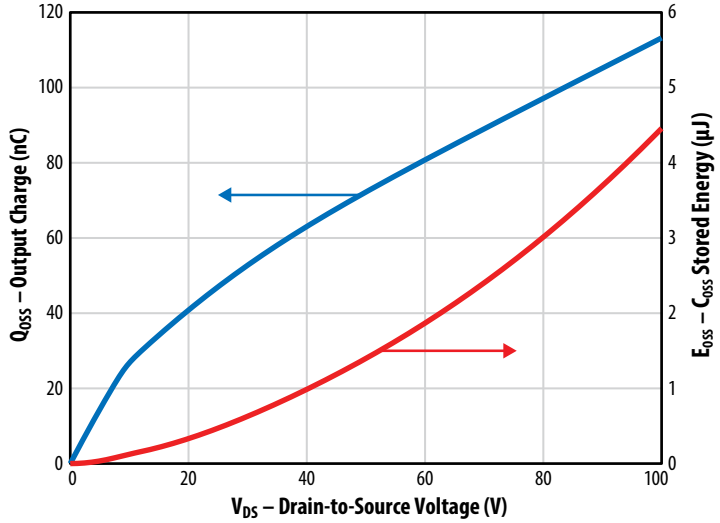


Figure 7: Typical Gate Charge

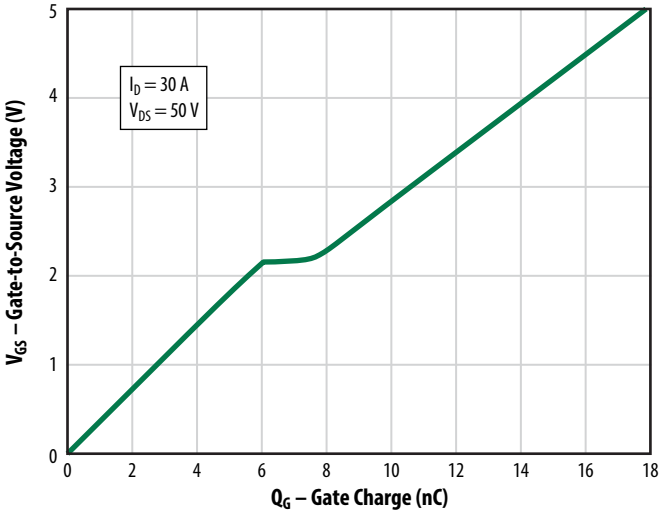
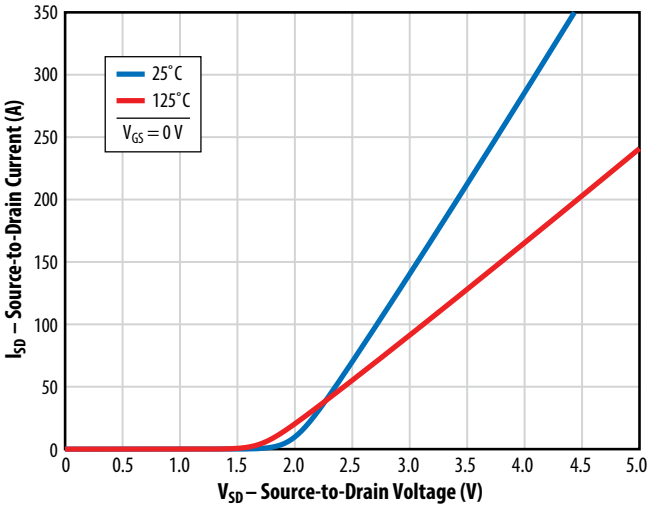


Figure 8: Reverse Drain-Source Characteristics



Note: Negative gate drive voltage increases the reverse drain-source voltage.
EPC recommends 0 V for OFF.

Figure 9: Normalized On-State Resistance vs. Temperature

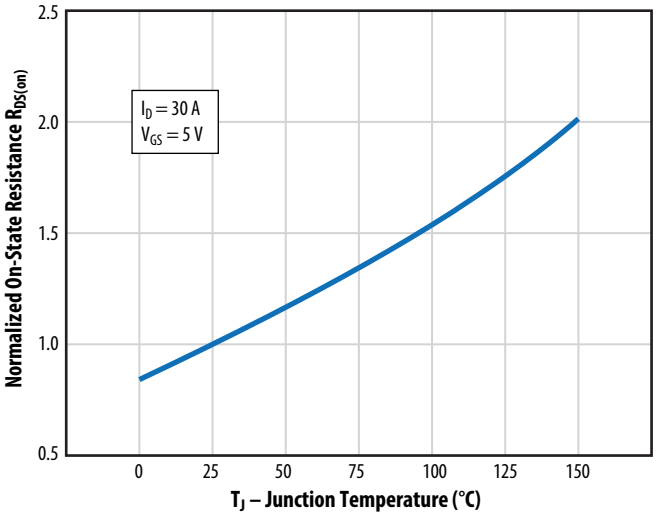


Figure 10: Normalized Threshold Voltage vs. Temperature

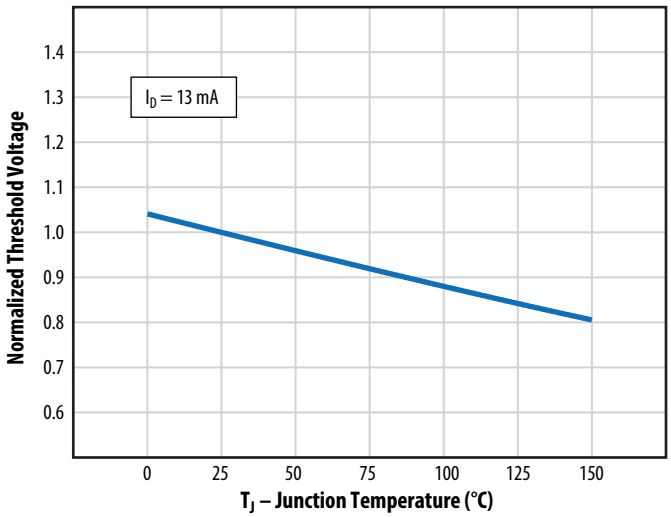
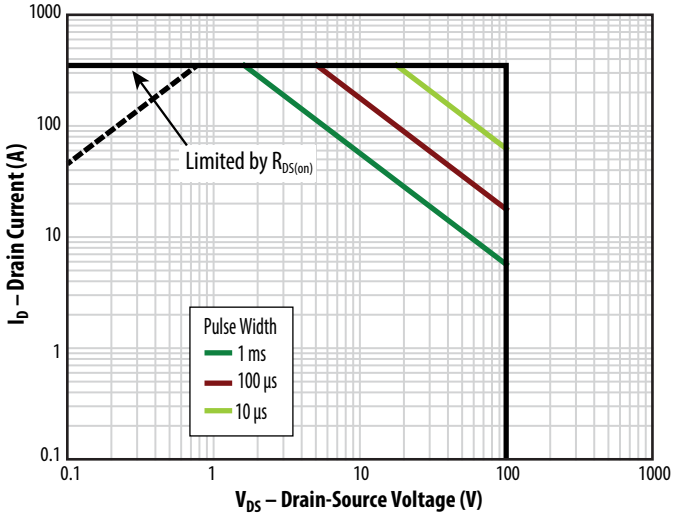


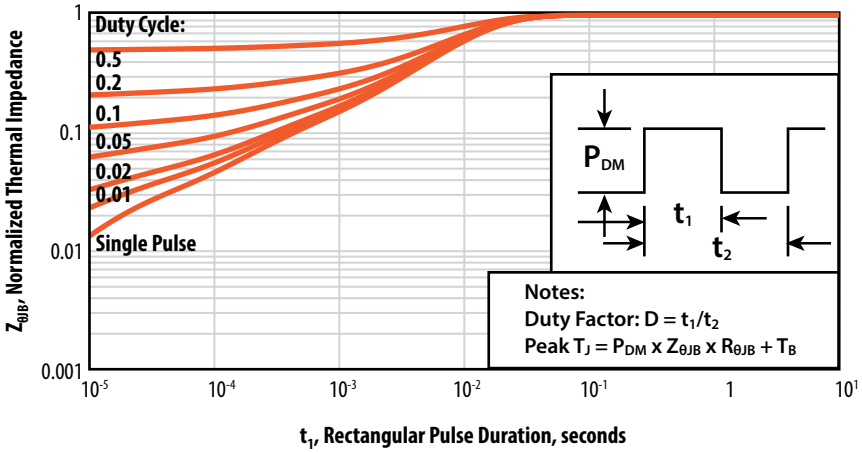
Figure 11: Safe Operating Area



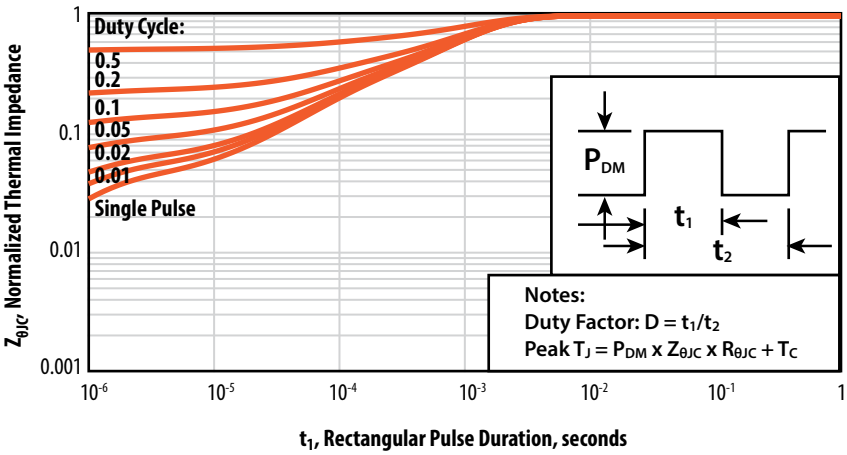
T_J = Max Rated, T_C = +25°C, Single Pulse

Figure 12: Transient Thermal Response Curves

Junction-to-Board

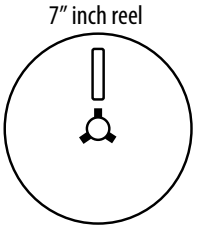


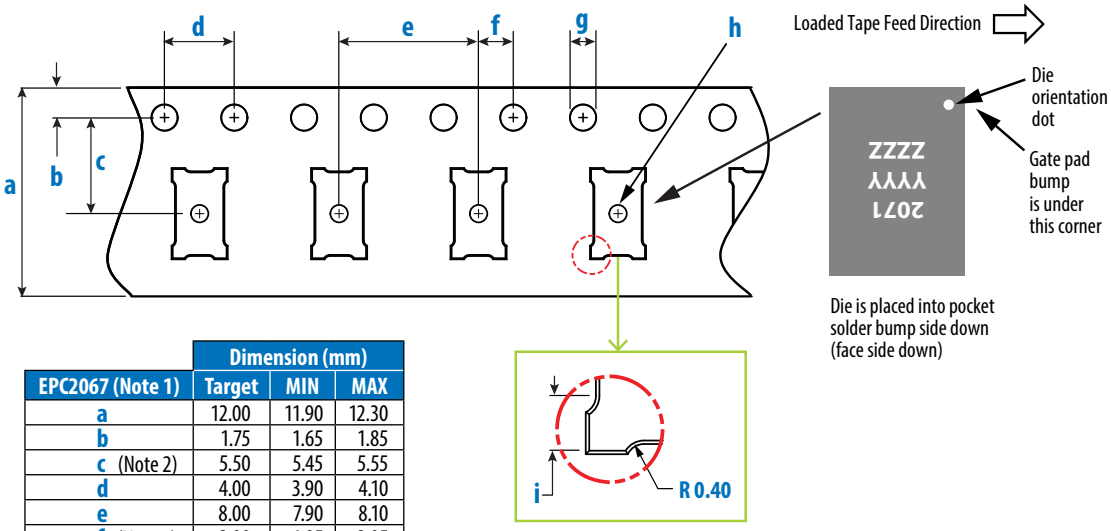
Junction-to-Case



TAPE AND REEL CONFIGURATION

8 mm pitch, 12 mm wide tape on 7" reel



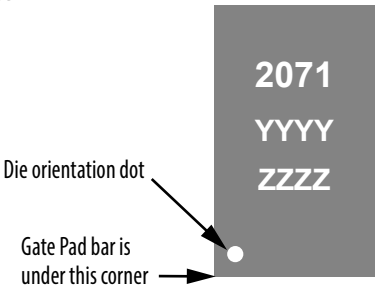


EPC2067 (Note 1)	Dimension (mm) Target	MIN	MAX
a	12.00	11.90	12.30
b	1.75	1.65	1.85
c (Note 2)	5.50	5.45	5.55
d	4.00	3.90	4.10
e	8.00	7.90	8.10
f (Note 2)	2.00	1.95	2.05
g	1.50	1.50	1.60
h	1.00	0.95	1.05
i	1.02		

Note 1: MSL 1 (moisture sensitivity level 1) classified according to IPC/JEDEC industry standard.

Note 2: Pocket position is relative to the sprocket hole measured as true position of the pocket, not the pocket hole.

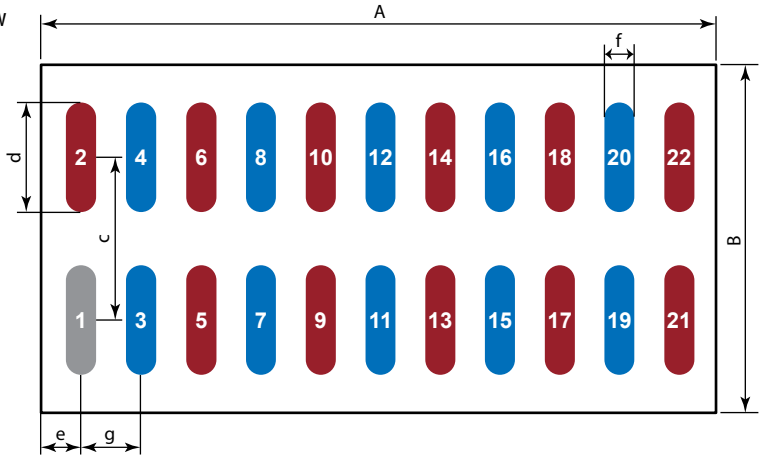
DIE MARKINGS



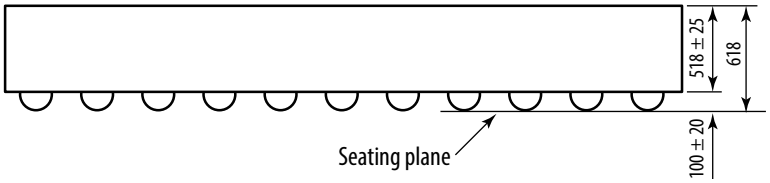
Part Number	Laser Markings Part # Marking Line 1	Lot_Date Code Marking Line 2	Lot_Date Code Marking Line 3
EPC2071	2071	YYYY	ZZZZ

DIE OUTLINE

Solder Bump View



Side View



DIM	Micrometers MIN	Nominal	MAX
A	4420	4450	4480
B	2270	2300	2330
c		1330	
d		720	
e		225	
f		200	
g		400	

Pad 1 is Gate;

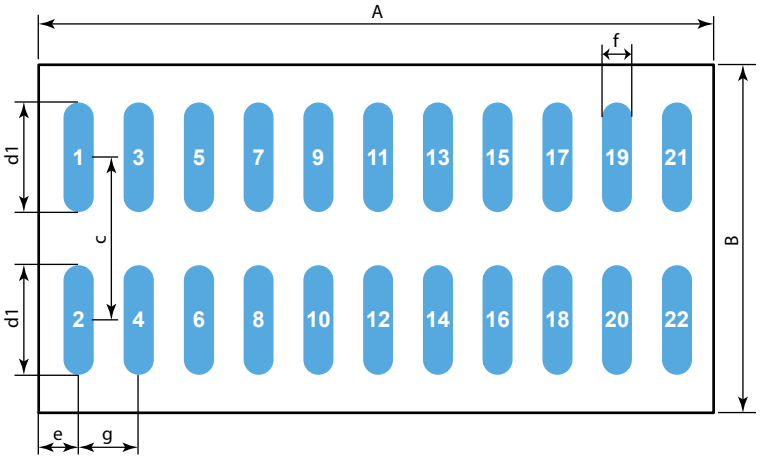
Pads 2 ,5, 6, 9, 10, 13, 14, 17, 18, 21, 22 are Source;

Pads 3, 4, 7, 8, 11, 12, 15, 16, 19, 20 are Drain;

*Substrate (top side) connected to Source

RECOMMENDED LAND PATTERN

(units in µm)

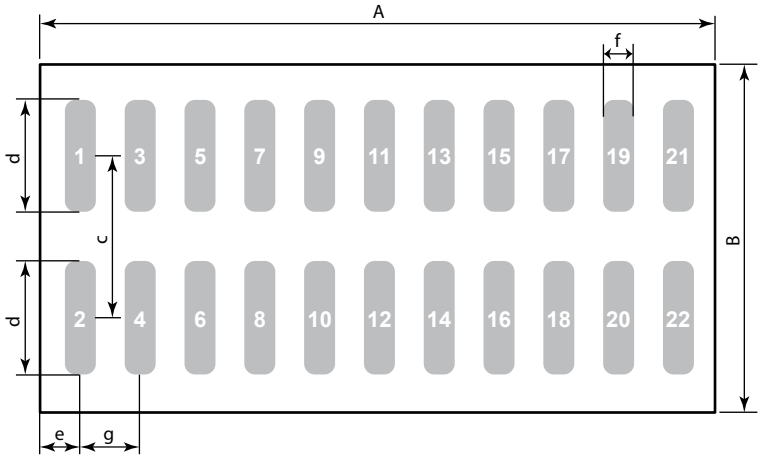


DIM	Micrometers
A	4450
B	2300
c	1330
d1	700
e	225
f	180
g	400

Pad 1 is Gate;
Pads 2, 5, 6, 9,10,13,14, 17, 18, 21, 22 are Source;
Pads 3, 4, 7, 8, 11, 12, 15, 16, 19, 20 are Drain

RECOMMENDED STENCIL DRAWING

(units in µm)



Recommended stencil should be 4 mil (100 µm) thick, must be laser cut, openings per drawing.

The corner has a radius of R60.

Intended for use with SAC305 Type 4 solder, reference 88.5% metals content.

DIM	Micrometers
A	4450
B	2300
c	1330
d	700
e	225
f	180
g	400

Additional resources available:

- Assembly resources – https://epc-co.com/epc/Portals/0/epc/documents/product-training/Appnote_GaNassembly.pdf
- Library of Altium footprints for production FETs and ICs – https://epc-co.com/epc/documents/altium-files/EPC%20Altium%20Library.zip (for preliminary device Altium footprints, contact EPC)

Efficient Power Conversion Corporation (EPC) reserves the right to make changes without further notice to any products herein to improve reliability, function or design. EPC does not assume any liability arising out of the application or use of any product or circuit described herein; neither does it convey any license under its patent rights, nor the rights of others.

eGaN® is a registered trademark of Efficient Power Conversion Corporation.
EPC Patent Listing: epc-co.com/epc/AboutEPC/Patents.aspx

Information subject to change without notice.
Revised June, 2022